

RAFEEQ SmartWheel

Mechanical Subsystem — Presentation Slide Brief
Version 2 · Complete Edition

Prepared for: Presentation Developer / HTML Engineer
CIE 598/599 · Team 06 · Zewail City of Science and Technology
Graduation Defense 2025/2026

How to use this document: Read Section 1 first for the big picture. Implement slides A→D in order. Exact text cells are verbatim for slides. File names in `monospace` must match exactly. References in Section 7 are for presenter’s speaking notes.

Slides covered	4 slides: A (Design Challenge), B (Motor Selection), C (5 Iterations), D (CAD + Manufacturing)
Insertion point	After existing <code>slide-9</code> in <code>rafeeq_final.html</code> . Total becomes 28 slides.
Estimated time	7–9 minutes total for the mechanical section
Video assets	<code>cad_motion_study.mp4</code> (50 sec) · <code>manufacturing_wirecut_video.mp4</code> (3 sec)
Image assets	13 images total — see Section 6 for all names and rename instructions
Brand colours	Navy #0D1F3C · Gold #C9992A · Gold-light #E8B84B · Red #C94040 · Green #2E7D52 · Amber #D97706
Font	Inter (already imported in <code>rafeeq_final.html</code> via Google Fonts)

CONTENTS

1. Overview — Section Structure & Story Arc

2. Slide A — The Design Challenge

Purpose, layout, exact text, animation sequence

3. Slide B — Motor Selection: Engineering Problem → Solution

Problem framing, torque sizing, 4-candidate card flip, motor images, spec strip, references

4. Slide C — 5 Iterations Timeline

All 6 states with exact content, photos, timeline behaviour, key interception

5. Slide D — CAD Video + Manufacturing Gallery

Video specs, 3-sec wire EDM clip, callout annotations, photo strip, tagline

6. Asset Reference — All File Names

Images, videos, rename instructions, folder structure

7. References & Sources

All citations used in motor selection and iteration design

8. Technical Implementation Notes

JS key interception, card flip CSS, video playback, colour tokens

SECTION 1

Overview — Section Structure & Story Arc

The Engineering Narrative

The Arc in One Sentence: “We faced three hard constraints, sized the motors through a quantitative torque analysis, chose the actuators through structured comparison, then designed the mounting system through five physical iterations — and manufactured it ourselves in Egyptian workshops.”

#	Name	Story Beat	Presenter	Time
A	The Design Challenge	Set up WHY this was hard — 3 constraints	Ziad	~30 s
B	Motor Selection	Problem statement → torque sizing → 4-candidate evaluation → selection	Ziad	~2 min
C	5 Iterations	Walk every physical iteration with real photos	Ziad + Nourhan	~4 min
D	CAD + Manufacturing	CAD motion study + workshop photos	Omar	~2 min

Insertion: Find `id="slide-9"` (dark "How We Built It" divider). Insert 4 new slides immediately after as `slide-10`, `-11`, `-12`, `-13`. Renumber all following slides +4. Update slide counter total: 24 → 28. Search JS for any hardcoded `totalSlides` and update.

SECTION 2

Slide A — The Design Challenge

Purpose

30-second setup slide. Frames the engineering problem before any solution is shown. **No images, no interaction, no clicks required.** Full navy background.

Exact Slide Content

Element	Exact Text	Style
Eye brow	MECHANICAL SUBSYSTEM	Gold, all-caps, letter-spaced, centred
Headline line 1	The Design	White, 48–56 px bold, left-aligned
Headline line 2	Challenge.	Same as above
Row 1	• Load — Support a 120 kg user on uneven indoor floors	Gold dot, bold label, white text
Row 2	• Constraint — Attach to a frame never designed for motor torque	Same pattern
Row 3	• Goal — Modular: install and remove without permanent modification	Same pattern
Closing line	“We didn’t design the final mount in a single pass. We iterated five times.”	Gold italic, centred, ~14 px

Animation Sequence

Step	Element	Animation	Delay
1	Eye brow	fadeDown	0 ms
2	Headline	fadeUp	200 ms
3	Row 1 (Load)	fadeRight (slide from left)	500 ms
4	Row 2 (Constraint)	fadeRight	900 ms
5	Row 3 (Goal)	fadeRight	1300 ms
6	Closing gold italic line	fadeIn	2000 ms

No images or videos required for Slide A.

SECTION 3

Slide B — Motor Selection: Engineering Problem → Solution

Purpose & Engineering Framing

This slide should feel like engineering thinking made visible. Structure: **Problem** → **Requirements** → **Torque Sizing** → **Candidates** → **Evaluation** → **Selection**. Not "here's what we chose" — "here's how we chose it".

Slide Opening — The Problem Statement (on screen 5 sec before cards appear)

Large navy centred text:

"We need to move a 120 kg loaded wheelchair reliably, at low speed, with precise independent wheel control — at a price that works in Egypt."

Smaller grey below:

Four motor types were evaluated against four criteria. Only one passed all of them.

Motor Sizing — What Goes on the Slide (Before the Cards)

Presenter note: This is a single small panel that appears *before* the 4 candidate cards. It takes **20–30 seconds** to present. One question, one equation, one answer. The committee sees you did the engineering — they do *not* need to follow every step. All spoken context is in the bullet points below the box.

The question: What is the minimum torque each motor must deliver to move a 120 kg loaded wheelchair up an ADA-compliant indoor ramp (1:12 gradient) without stalling?

$$T_{\text{required}} = \frac{mg \cdot \sin \theta \cdot r}{2} \times SF = \frac{120 \times 9.81 \times \frac{1}{12} \times 0.130}{2} \times 1.25 = \mathbf{7.9 \times 1.25 \approx 10 \text{ N} \cdot \text{m per motor}}$$

Selected motor delivers: $T_{\text{BLDC}} = 11 \text{ N} \cdot \text{m}$ at $I = 15 \text{ A} \Rightarrow 11 > 10$ **Passes. ✓**

$m = 120 \text{ kg}$ (user + chair) $\sin \theta = 1/12$ (ISO ADA ramp) $r = 0.130 \text{ m}$ (8-inch hub motor) $SF = 1.25$ (ISO 13482)

Presenter speaks (not on screen):

- "We sized the motors against the worst-case scenario: a 120 kg user on a 1-in-12 ramp."
- "The calculation gives us a requirement of 10 N · m per motor with a safety factor."
- "The selected BLDC hub motor delivers 11 N · m at our configured current limit — it passes with margin."
- "That result is confirmed in the firmware: `I_MOT_MAX = 15 A`, Thesis Table 4.8."
- *[transition]* "Now — how did we actually choose this motor over the alternatives?"

Evaluation Criteria (4 icons shown across top of slide)

Criterion	Icon	What it means	Why it matters
Reliability	•	No brushes, gears, or wear parts	Medical device — must not fail at home. Maintenance unacceptable.
Torque at low speed	!	$\geq 10 \text{ N} \cdot \text{m}$ at 0–1.5 m/s	ADA ramp 1:12 without stalling (ISO 7176-14, §3.2.1.3)
Closed-loop control	§	Position/speed feedback	Without feedback, straight-line drift uncorrectable in indoor nav
Cost	£	Fit within BOM budget	Medical wheelchair motors cost 3–5× more — Egyptian market

The 4 Candidate Cards — Complete Content

Card 1 × REJECTED

Motor: Brushed DC Motor

Image file: motor_brushed_dc.jpg — Generic brushed DC motor, visible commutator and brushes. Source: web product image.

Criterion	Pass?	Reasoning
Reliability	×	Brushes wear after 6–12 months of continuous use. Replacement required — unacceptable for a home medical device.
Torque	✓	Adequate torque, but performance degrades as brushes wear.
Closed-loop control	△	Can use external encoder, but no built-in Hall sensor. Adds cost and complexity.
Cost	✓	Low initial cost — but high lifetime maintenance cost.

Ref: Krishnan, R. — *Electric Motor Drives*, Prentice Hall, 2001 [58]

Card 2 × REJECTED

Motor: Stepper Motor

Image file: motor_stepper.jpg — NEMA-17 or NEMA-23 stepper. Source: web product image.

Criterion	Pass?	Reasoning
Reliability	✓	No brushes. Mechanically robust.
Torque	×	Open-loop by design — loses steps under dynamic load. Cannot confirm actual wheel position. Torque drops significantly at speed.
Closed-loop control	×	No native speed feedback. External encoder required for closed-loop — complexity defeats the purpose.
Cost	✓	Moderate cost.

Ref: Krishnan, R. — *Electric Motor Drives* [58]

Card 3 × REJECTED

Motor: Servo Motor + Gearbox

Image file: motor_servo_gearbox.jpg — Industrial servo with gearbox. Source: web product image.

Criterion	Pass?	Reasoning
Reliability	△	Motor is reliable, but gearbox introduces backlash, wear, and additional failure points.
Torque	✓	Gearbox multiplies torque — good low-speed performance.
Closed-loop control	✓	Closed-loop encoder. Excellent control.
Cost	×	Most expensive option: servo driver + gearbox + motor = 3–4× cost of hub motor solution.

Ref: Cooper et al. — *Evaluation of electric-powered wheelchairs*, APMR 1999 [49]

Card 4 ✓ SELECTED

Motor: BLDC Hub Motor (8-inch, 350 W)

Image file: motor_bldc_hub.jpg — YOUR actual hoverboard 8-inch hub motor. Photo of the component used.

Criterion	Pass?	Reasoning
Reliability	✓	No brushes, no gears, no belts. Direct-drive. Zero wear parts in drivetrain.
Torque	✓	FOC delivers max torque at zero speed. $11 \text{ N} \cdot \text{m}$ at $15 \text{ A} > 10 \text{ N} \cdot \text{m}$ required. ADA ramp validated.
Closed-loop control	✓	Built-in Hall effect sensors — 3 sensors, 6 transitions/rev. Closed-loop FOC via EFeru STM32 firmware.
Cost	✓	$\approx 70\%$ cheaper than dedicated medical wheelchair motors. Leverages hoverboard mass-production supply chain.

Ref: Krishnan [58] · Åström & Hägglund [57] · ISO 7176-14 [31] · EFeru firmware [37]

Spec Strip (appears after all 4 cards are flipped)

Wheel diameter	Power	Supply voltage	Control	Sensing
8 in (203 mm)	$350 \text{ W} \times 2$	36 V Li-ion 10S2P	FOC · EFeru STM32	Hall effect · 3 sensors/motor

SECTION 4

Slide C — 5 Iterations Timeline

Purpose

Walk the committee through the iterative design process using real photos and engineering language. **The most important slide for demonstrating rigour.** Should feel like turning pages of an engineering journal.

Fixed Top Bar (does not change between iterations)

Eyebrow: ITERATIVE DESIGN PROCESS · MECHANICAL SUBSYSTEM

Headline: Five Iterations. One Validated Mount.

Navigation Behaviour

- **Inside Slide C:** Pressing → or clicking a timeline dot advances to next iteration. Photo + text crossfade (300 ms). Active dot slides right.
- **Exit forward:** Pressing → on FINAL state moves to Slide D.
- **Exit backward:** Pressing ← on Iteration 1 goes back to Slide B.
- See Section 8 for JS key interception pattern.

Layout Per State

Left panel (55%): full-height photo, `object-fit: cover`, small iteration badge overlay top-left, tag bottom-left. Right panel (45%): iteration number + verdict badge, name, then: “What we tried” / “What happened” / “What we learned”. Bottom: fixed timeline bar, 6 dots.

Iteration 1 × REJECTED Bent-plate bracket

`iter1_prototype.jpg` Tag: ** Bracket Design · Phase 1*

What we tried: A simple bent-plate bracket to clamp the motor housing directly to the wheelchair’s side frame tube.

What happened: Hole spacing and bend angles didn’t match the motor housing or frame tube diameter. Motor visibly misaligned. Assembly unstable under hand load.

What we learned: A single bent plate is insufficient — bracket must be a **closed, self-referencing structure**. Motivated redesign into box-section.

Iteration 2 × REJECTED Stainless steel box-section bracket

`iter2_steel_box.jpg` Tag: ** Bracket Design · Phase 1*

What we tried: Welded stainless steel box section wrapping the frame tube on 3 sides, clamped with through-bolts. Corrects alignment issue of Iter 1.

What happened: Under bolt torque, thin walls bowed inward — distorting section geometry and loosening the clamp over repeated tightening cycles.

What we learned: **Wall thickness and bolt-bearing area** are the limiting factors, not bracket shape. Switched to solid flat-bar stock.

Iteration 3 ! PARTIAL Mild steel flat-bar bracket with rigid cross-bar

`iter3_flatbar_bracket.jpg` Tag: ** Bracket Design · Phase 1*

What we tried: Solid mild steel flat bar, wider bolt pattern, two motor brackets tied by a single rigid cross-bar.

What happened: ✓ **Zero deflection under 120 kg static load. Structurally validated.** BUT: rigid cross-bar caused one wheel to lift on uneven floors.

What we learned: Bracket geometry is now correct. More fundamental system problem exposed: **the two drive wheels must be mechanically independent.**

Iteration 4 ✓CAD VALIDATED Independent suspension redesign (CAD)

iter4_cad_suspension_concept.png Tag: [CAD] CAD Design · Phase 2

What we tried: Complete redesign — two fully independent scissor/pantograph suspension links. Each motor on its own bracket. Each wheel can move vertically independent of the other.

What happened: CAD validation confirmed geometry. Each wheel maintains independent floor contact. Gas struts provide constant downforce. 120 kg rated.

What we learned: Independent suspension concept is correct. Design sent to manufacturing: CNC milling + Wire EDM.

Iteration 5 !FIXED Manufactured mount — alignment correction

iter5_problem_misalignment.jpg Tag: [MFG] Manufacturing · Phase 3

What we tried: Iteration 4 design manufactured (CNC milled brackets + Wire EDM) and installed on both sides of the wheelchair.

What happened: Two geometric issues found: (1) **Wheel camber** — one drive wheel leaned, contacting floor only at its edge. (2) **Toe-in** — front edges of both wheels closer together, producing constant rolling friction. Both traced to tolerance stack-up.

What we learned: Brackets re-machined and re-aligned to zero camber and zero toe relative to chair centreline.

Final Configuration ✓VALIDATED Re-aligned brackets

final_configuration.jpg Tag: ✓Production-Ready

What we tried: Re-machined and re-aligned brackets reinstalled on both sides.

What happened: **Zero camber. Zero toe. Both drive wheels achieve full-width floor contact.** Independent suspension maintains contact on uneven surfaces. 120 kg: zero measurable deflection. No rotational slip.

What we learned: Final mount configuration validated (Thesis Table 4.2, §4.2.7). Modular installation confirmed. No permanent modification to wheelchair frame.

Timeline Dot Legend

State	Colour	Behaviour
Active (current)	Gold #C9992A, 16 px	Large filled dot with subtle glow
Completed	Gold #C9992A, 10 px	Smaller, no glow
Upcoming	Grey #9CA3AF, 10 px	Hollow or faded
FINAL dot	Green #2E7D52 when active	Switches to gold when complete
Connecting line	Grey #E5E7EB, 1 px	Full width at fixed position

SECTION 5

Slide D — CAD Video + Manufacturing Gallery

Two messages on one slide: (1) CAD motion study = designed with rigour. (2) Manufacturing photos + wire EDM video = built for real, in Egyptian workshops. Together: *production-readiness signal*.

Moment 1 — CAD Motion Study Video (top 58% of slide)

Property	Value
File name	<code>cad_motion_study.mp4</code>
Duration	~50 sec original → plays at 1.5× speed = ~33 sec displayed
Placement	Centred horizontally, top 58% of slide. Rounded corners 12 px. Navy border 1.5 px.
Autoplay	NO. Custom gold play button overlay. Presenter clicks when ready.
Speed	<code>video.playbackRate = 1.5</code> triggered immediately on play. No controls bar visible.
After video ends	3 callout annotations fade in (staggered, see below).

#	Position	Icon	Text on Slide	Delay after end
1	Left of video	[M]	Modular plug-in interface — no permanent modification to the wheelchair	0 ms
2	Below video centre	[S]	Scissor/pantograph linkage with gas struts — constant down-force on drive wheels	300 ms
3	Right of video	[W]	120 kg rated — each wheel independently follows the floor	600 ms

Moment 2 — Manufacturing Gallery (bottom 38% of slide)

Position	Type	File Name	Caption	Lightbox	Delay
Left (1/3)	Photo	<code>manufacturing_cnc.jpg</code>	CNC Milling	Open fullscreen	500 ms
Centre (2/3)	Video clip	<code>manufacturing_wirecut_video.mp4</code>	Wire EDM	Autoplay inline, loops twice	700 ms
Right (3/3)	Photo	<code>final_configuration.jpg</code>	Final Assembly	Open fullscreen	900 ms

Wire EDM video (3 sec): Autoplays when thumbnail appears. Set to loop twice (≈6 sec total). Listen for ended event, increment loop counter, replay if count < 2.

Bottom tagline (centred, navy italic, fadeIn 1.2 s after photo strip):
“Designed in Egypt. Machined in Egypt. Built for Egypt.”

SECTION 6

Asset Reference — All File Names

Folder Structure

```

rafeeq_final.html
assets/mech/
iter1_prototype.jpg
iter2_steel_box.jpg
iter3_flatbar_bracket.jpg
iter4_cad_suspension_concept.png
iter5_problem_misalignment.jpg
final_configuration.jpg
manufacturing_cnc.jpg
manufacturing_wirecut.jpg
motor_brushed_dc.jpg
motor_stepper.jpg
motor_servo_gearbox.jpg
motor_blcdc_hub.jpg
cad_motion_study.mp4
manufacturing_wirecut_video.mp4

```

Complete Image List

File Name	Used In	Description	Action
iter1_prototype.jpg	Slide C · Iter 1	First prototype — old chair, aluminium extrusion with two vertical tubes	Already have
iter2_steel_box.jpg	Slide C · Iter 2	Stainless steel box-section bracket on black surface	Already have
iter3_flatbar_bracket.jpg	Slide C · Iter 3	Polished mild-steel flat bar T-bracket with cross-bar rod	Already have
iter4_cad_suspension_concept.png	Slide C · Iter 4	CAD render — two independent yellow BLDC hub motors, scissor linkage	Already have
iter5_problem_misalignment	Slide C · Iter 5	Final chair on tiled floor — visible wheel camber/toe before correction	Already have
final_configuration.jpg	Slide C · Final & Slide D	Clean final chair rear view — both hub motors flat, gas struts	Already have
manufacturing_cnc.jpg	Slide D · Photo strip	CNC milling machine mid-operation, metal chips visible	Rename: 1781449010835_image.
manufacturing_wirecut.jpg	Slide D · Thumbnail	Wire EDM machine (blue frame), water bath visible	Rename: 1781449022102_image.
motor_brushed_dc.jpg	Slide B · Card 1	Generic brushed DC motor, commutator visible	Find online
motor_stepper.jpg	Slide B · Card 2	NEMA-17 or NEMA-23 stepper motor	Find online
motor_servo_gearbox.jpg	Slide B · Card 3	Servo motor with gearbox	Find online
motor_blcdc_hub.jpg	Slide B · Card 4	YOUR 8-inch BLDC hoverboard hub motor	Photograph it

Video Assets

File Name	Used In	Specs
cad_motion_study.mp4	Slide D · Top zone	~50 sec · 1.5× playback · MP4 H.264 · Local host · No autoplay
manufacturing_wirecut_video.mp4	Slide D · Centre strip	3 sec · Normal speed · Loops ×2 · MP4 H.264 · Local host

SECTION 7

References & Sources

For presenter's speaking notes. Most are not shown on screen; cite verbally when the committee asks.

Motor Selection References

- [58] R. Krishnan, *Electric Motor Drives: Modeling, Analysis, and Control*. Prentice Hall, 2001. → Core reference for BLDC motor operating principles, FOC theory, and motor type comparison.
- [57] K. J. Åström & T. Hägglund, *Advanced PID Control*. ISA, 2006. → PID controller theory underpinning FOC closed-loop control cited in criteria.
- [31] ISO 7176-14:2008, *Wheelchairs — Part 14: Power and control systems for electrically powered wheelchairs*. ISO, Geneva, 2008. → ADA ramp compliance standard (1:12 slope) used in torque sizing calculation.
- [33] ISO 13482:2014, *Robots and robotic devices — Safety requirements for personal care robots*. ISO, Geneva, 2014. → Safety factor 1.25 used in torque sizing derivation.
- [37] EFeru, *Hoverboard Firmware Hack FOC*. GitHub, 2023. <https://github.com/EFeru/hoverboard-firmware-hack-1> → FOC firmware on STM32F103RCT6. Source for 11 N·m at 15 A figure.
- [49] R. A. Cooper, M. L. Boninger & A. Rentschler, "Evaluation of selected electric-powered wheelchairs using the ANSI/RESNA standards," *Archives of Physical Medicine and Rehabilitation*, vol. 80, no. 6, pp. 693–702, 1999. → Servo/gearbox rejection context; medical wheelchair standards.
- [40] P. Vas, *Vector Control of AC Machines*. Oxford University Press, 1990. → FOC / vector control theoretical basis.

Mechanical Design References

- [29] ISO 7176-4:2008, *Wheelchairs — Part 4: Energy consumption of electric wheelchairs*. ISO, 2008.
 - [46] ISO 7176-14:2022, *Wheelchairs — Power and control systems*. ISO, 2022. → Speed compliance: $N_{MOT_MAX} = 90$ RPM.
 - [47] IEC 60364-5-52:2009, *Low-voltage electrical installations — Wiring systems*. IEC, 2009.
- Thesis Team 06, "Mechanical Subsystem — Iterative Design Process," *GRADUATION_THEEESIS_Final_Version_2026*, pp. 77–84, 2026. → Primary source for all 5 iteration descriptions and test results.

SECTION 8

Technical Implementation Notes

8.1 Slide Insertion in rafeeq_final.html

Find `id="slide-9"` (dark divider). Insert 4 new slides after it: `slide-10`, `slide-11`, `slide-12`, `slide-13`. Renumber existing: `10→14`, `11→15`, ..., `24→28`. Update `#slide-counter` to `"/28"`. Search JS for any hardcoded `"24"` and update.

8.2 Slide C Key Interception Logic

1. When Slide C becomes active: `window.inIterationMode = true; window.iterIndex = 0;`
2. In existing keydown handler: check if `(window.inIterationMode)` before calling next/prev slide.
3. `ArrowRight` → call `advanceIteration()`: increments `iterIndex`. If `iterIndex > 5`: set flag false, call `nextSlide()`.
4. `ArrowLeft` → call `prevIteration()`: decrements. If `iterIndex < 0`: set flag false, call `prevSlide()`.
5. Dot clicks: `goToIteration(n)` sets index directly.
6. On leaving Slide C: `window.inIterationMode = false; window.iterIndex = 0;`

8.3 CSS Card Flip (Slide B)

Container has class `flip-card` with `perspective: 1000px`. Inner div has class `flip-inner` with `transform-style: preserve-3d` and `transition: transform 0.55s cubic-bezier(0.16, 1, 0.3, 1)`. Two children: `flip-front` and `flip-back`, absolutely positioned with `backface-visibility: hidden`. The back starts at `transform: rotateY(180deg)`. Toggle class `flipped` on click. Show spec strip when all 4 cards have class `flipped`.

8.4 Video 1.5× Speed + Callout Trigger

On play button click: call `vid.play()` then immediately set `vid.playbackRate = 1.5`. Hide the play button overlay. On the ended event: use `setTimeout` to show callout 1 at 0 ms, callout 2 at 300 ms, callout 3 at 600 ms, manufacturing strip at 1100 ms, tagline at 1900 ms.

8.5 Wire EDM Video — Auto-loop-twice

Listen to `ended` event on `wirecut-video` element. Keep a counter. On first end (`count = 1`), call `play()` again. On second end (`count = 2`), stop. Gives ≈6 seconds of viewing from a 3-second clip.

8.6 Brand Colour Tokens

CSS Variable	Hex	Use
<code>-navy</code>	<code>#0D1F3C</code>	Dark slides, headers, nav bar, card fronts
<code>-teal / -gold</code>	<code>#C9992A</code>	All gold accents, timeline dots, borders, active badges
<code>-teal-light</code>	<code>#E8B84B</code>	Hover states, glow, spec strip text, eyebrow letters
<code>-coral / red</code>	<code>#C94040</code>	REJECTED badges, rejected card backs, error states
<code>green</code>	<code>#2E7D52</code>	SELECTED / VALIDATED badges, FINAL iteration dot
<code>amber</code>	<code>#D97706</code>	PARTIAL / FIXED badges (Iterations 3 and 5)
<code>-muted / gray</code>	<code>#9CA3AF</code>	Inactive timeline dots, secondary text, captions